BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

Paper II (Objective Type) — Inter-A-2021

پرچہ II (معروضی)

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8197

Q1. Lim x->3 sqrt(x2 + x + 4) =:

(A) 4
(B) -4
(C) 6
(D) 0

Q2. d/dx (Sin-1x) =:

(A) -1/sqrt(1-x2)
(B) 1/(1+x2)
(C) -1/(1+x2)
(D) 1/sqrt(1-x2)

Q3. d/dx (Sinx) =:

(A) -Cos x
(B) Cos x
(C) tan x
(D) Sec x

Q4. Lim x->0 Sin(ax)/Sin(bx) =:

(A) b/a
(B) 1
(C) a/b
(D) -a/b

Q5. 1 - x2/2! + x4/4! - x6/6! + ----- is the Maclaurin's Series Expansion of:

(A) Sin x
(B) Cos x
(C) ex
(D) e2x

Q6. ∫ 1/x dx:

(A) x-2/-2 + C
(B) x0/0 + C
(C) 1/x2 + C
(D) ln x + C

Q7. ∫ tan2x dx =:

(A) 2 tan x + c
(B) tan x - c
(C) tan x + x + c
(D) tan x - x + c

Q8. If f(x) = ax then f'(x) =:

(A) ax lna
(B) ax
(C) x ax-1
(D) ax/lna

Q9. ∫ Cosec x dx:

(A) -Cosec x Cot x + C
(B) ln(Cosec x - Cot x) + C
(C) ln(Cosec x + Cot x) + C
(D) ln(Sec x + tan x) + C

Q10. Slope of line 2y = x - 7 is:

(A) 1/2
(B) 2
(C) -1/2
(D) -2

Q11. The distance of the point (2,3) from y-axis is:

(A) 3
(B) 2
(C) -2
(D) -3

Q12. ∫ ex(Cosx + Sinx) dx:

(A) ex Sinx + C
(B) ex Cosx + C
(C) -ex Cosx + C
(D) -ex Sinx + C

Q13. Point Slope form of equation of Straight line is:

(A) y = mx + c
(B) y - y1 = m(x - x1)
(C) x/a + y/b = 1
(D) x Cosα + y Sinα = P

Q14. k^ x j^ =:

(A) i
(B) -i
(C) 1
(D) 0

Q15. If u = 2i - j + 4k then u x v =:

(A) 8
(B) 1
(C) -1
(D) 0

Q16. Which of the following is the solution of inequality x + 2y < 6:

(A) (4,1)
(B) (1,3)
(C) (3,3)
(D) (1,4)

Q17. If v = i - j - k then |v| =:

(A) sqrt(3) (B) 3

(C) 9 (D) 1

Q18. The equation x2/a2 - y2/b2 = 1 represents:

(A) Hyperbola (B) Circle

(C) Parabola (D) Ellipse

Q19. The Conic is a Hyperbola if Eccentricity is:

(A) e = 1 (B) 0 < e < 1

(C) e > 1 (D) e = 1/2

Q20. Directrix of the Parabola x2 = -4ay is:

(A) x = a (B) x = -a

(C) y = a (D) y = -a

Mathematics (Subjective) — Inter-A-2021

ریاضی (انشائیہ)

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: It is compulsory to attempt any (8-8) Parts each from Q.No. 2 to Q.No. 4. 25 x 2 = 50

Note: Part - I — 25 x 2 = 50

QQ.No.2.

(i) Given f(x) = x3 - ax2 + bx + 1, if f(2) = -3 and f(-1) = 0 find the values of a and b.

(ii) Evaluate Lim x->3 (x-3)/(sqrt(x)-sqrt(3))

(iii) Evaluate Lim x->0 (1-Cosx)/Sin2x

(iv) Find dy/dx if y = x/lnx

(v) Find y2 if y = sqrt(x) + 1/sqrt(x)

(vi) Prove that d/dx Sec-1x = 1/(|x| sqrt(x2-1))

(vii) Differentiate (x2+1)/(x2-3)

(viii) Find f'(x) if f(x) = ln(sqrt(e2x + e-2x))

(ix) For real valued function f, find f-1(-1), f(x) = (2x+1)/(x-1), x > 1

(x) Differentiate w.r.t. 'x' Sin-1 sqrt(1-x2)

(xi) Differentiate w.r.t. 'x': y = e-2x Sin2x

(xii) If y = x4 + 2x2 + 2 prove that dy/dx = 4x sqrt(y-1)

QQ.No.3.

(i) Find δy of y = f(x) = x2, when x = 2 and dx = 0.01

(ii) Find the area between the x-axis and the curve y = x2 + 1 from x = 1 to x = 2

(iii) Show that the points A(0,2), B(sqrt(3),-1) and C(0,-2) are vertices of right triangle.

(iv) Check whether the point (5,8) lies above or below the line 2x-3y+6 = 0

(v) Evaluate ∫ ex/(ex+3) dx

(vi) Evaluate ∫ Sin2x dx

(vii) Evaluate ∫ x lnx dx

(viii) Evaluate ∫(1 to 2) (x2 + 1)dx

(ix) Evaluate ∫ (2x+3)1/2 dx

(x) Solve the Differential Equation dy/dx = (1-x)/y

(xi) Find an equation of vertical line through (-5,3)

(xii) Find the equation of lines represented by 3x2 + 7xy + 2y2 = 0

QQ.No.4.

(i) Graph the Solution Set of 2x + y ≤ 6

(ii) Find the Centre and Foci of x2/4 - y2/9 = 1

(iii) Find the Centre and Radius of Circle 5x2 + 5y2 + 14x + 12y - 10 = 0

(iv) Write the equation of Tangent and Normal to the circle x2 + y2 = 25 at (4,3)

(v) Find the Focus and Vertex of the Parabola x2 = -16y

(vi) Find point of Intersection of Conics x2 + y2 = 8 and x2 - y2 = 1

(vii) Find equation of Parabola if Focus is (-3,1) and Directrix x = 3

(viii) Find a Unit Vector in the Direction of v = (1/2)i + (sqrt(3)/2)j

(ix) Find α, so that |αi + (α+1)j + 2k| = 3

(x) Find Cosine of angle between the vectors u = [2,-3,1] and v = [2,4,1]

(xi) Find Area of Parallelogram with vertices A(1,2,-1), B(4,2,-3), C(6,-5,2) and D(9,-5,0)

(xii) Find α if i - j + k, i - 2j - 3k and 3i - αj + 5k are Coplanar.

(xiii) Find the Direction Cosines of v = 3i - j + 2k

Note: Part - II

QQ.No.5. (Marks 5) (a) Find values of m and n so that f(x) = { mx if x < 3; n if x = 3; -2x+9 if x > 3 } is continuous at x = 3

Q. (Marks 5) (b) If tan y (1 + tan x) = 1 - tan x, then show that dy/dx = -1

QQ.No.6. (Marks 5) (a) Find ∫ Tan3x.Secx dx

Q. (Marks 5) (b) Find h such that the points A(h,1), B(2,7) and C(-6,-7) are the vertices of a Right Triangle with Right Angle at the Vertex A.

QQ.No.7. (Marks 5) (a) Evaluate ∫(-1 to 2) (x + |x|) dx

Q. (Marks 5) (b) Minimize z = 3x + y subject to constraints 3x + 5y ≥ 15, x + 6y ≥ 9, x ≥ 0, y ≥ 0

QQ.No.8. (Marks 5) (a) Show that the Circles x2 + y2 + 2x - 2y - 7 = 0 and x2 + y2 - 6x + 4y + 9 = 0 touch externally.

Q. (Marks 5) (b) Use Vectors Method, prove that in any Triangle ABC, a2 = b2 + c2 - 2bcCosA

QQ.No.9. (Marks 5) (a) If y = a Cos(lnx) + b Sin(lnx) prove that x2 d2y/dx2 + x dy/dx + y = 0

Q. (Marks 5) (b) A Parabolic Arch has a 100 m base and height 25 m. Find the height of the arch at the point 30 m from the centre of the base.